Hyak Klone Slurm Workshop



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INFORMATION TECHNOLOGY
UNIVERSITY of WASHINGTON



Agenda

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Introduction

UWIT Research Computing

- Services: Hyak Klone and Tillicum
- New to Hyak?
- User support
- Research Computing Club

Learning Resources

- Documentation
- Events

Tutorial

- Objectives
- Layout
- Hands-on option

Research Computing Services



- High-Performance Computing
  - Hyak- UW's Supercomputer
  - Tillicum - GPU cluster
- Data storage
  - Kopah Object Storage
  - Lolo Archive
- Research Cloud Computing
  - AWS, GCP, Azure
- Secure Computing
- Research Computing Consulting

Hyak Klone



- **UW's Supercomputer**
  - **HPC Cluster**
  - **Condo Model**
  - **Storage**
  - **Community Idle Resources**

  - **37,076 compute cores**
  - **864 GPUs**

Tillicum

LAUNCHED
Oct 15
2025

University of Washington's next-generation, GPU-accelerated computing platform designed to support cutting-edge research, teaching, and learning in AI, machine learning, data science, and scientific simulation.

WHERE PERFORMANCE MEETS SUSTAINABILITY

RANKED #7	RANKED #185	RANKED #51
FASTEST US HIGHER ED	FASTEST WORLDWIDE	POWER EFFICIENCY

WHY TILLICUM MATTERS?

Broad Academic Access

- 3 Campuses
- 11 Colleges & Schools
- 33 Departments
- 77 Research Groups
- 542 Users

Proven Demand & Impact

- 94,000 GPU Hours Consumed

SUPPORTING ALL DOMAINS

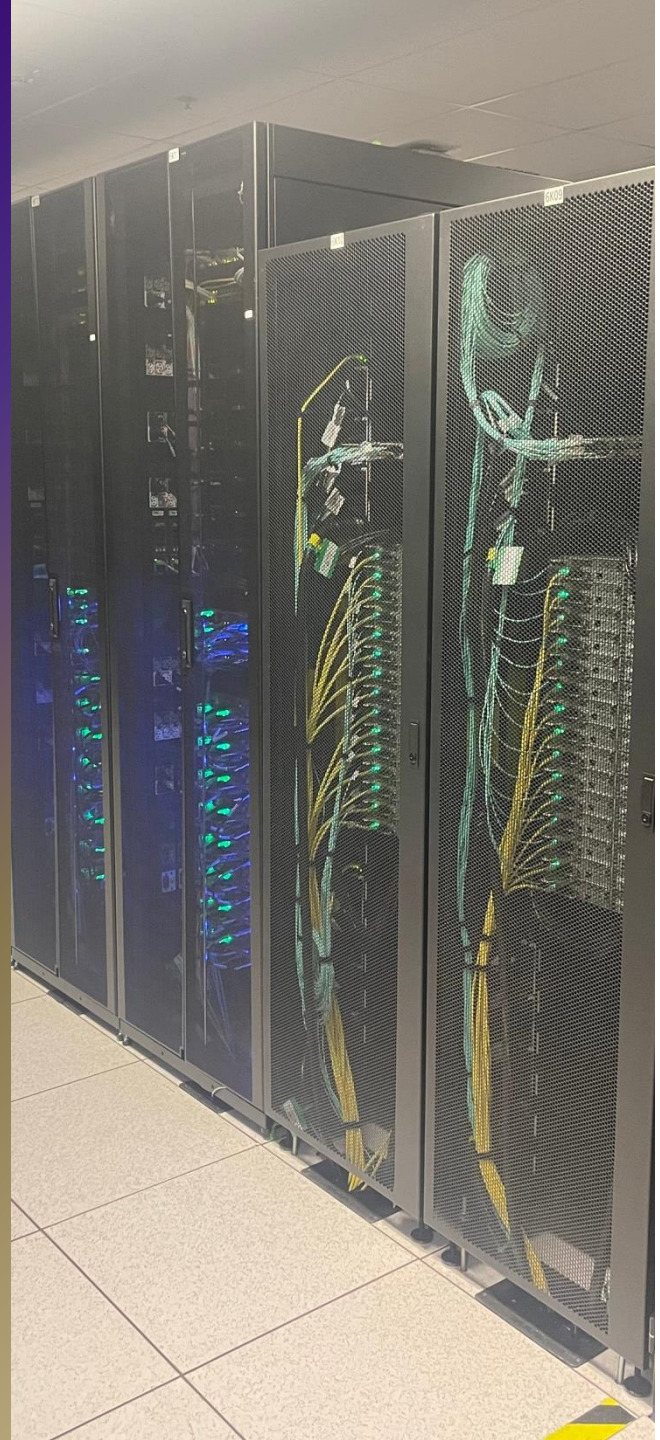
- Engineering
- Math
- Computer Science
- Physical & Life Sciences
- Quantum
- Teaching & Learning
- AI
- Medicine

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NVIDIA

DELL

[re]DESIGN



TILLICUM

- Pay-as-you-go subscription
- Usage limit enforcement control
- 24 Dell XE9680 Servers
  - 2TB RAM per node
- **192 NVIDIA H200** SXM GPUs (141GB RAM each)
- 1,536 CPU cores [Intel Emerald Rapids]
- NVLink™ 4.0 (900 GB/s)
- 400 Gbps InfiniBand
- 3 PB Flash Storage

Top500 benchmark gave us ~8.8 PFLOPs.
Green500 benchmark gave us ~46 GFLOPs/Watt.



NEW TO HYAK?

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Need-to-knows

- Klone is a paid service and a community resource
- Klone is run by a small but mighty team
  - We value our relationship with you

- Request support from our **Support Request form** or by emailing help@uw.edu with “Hyak” in the subject line.

- Hyak Klone and Tillicum are offline the second Tuesday of each month for regular maintenance.
- Reminders are sent via email to all users with access.

- Demonstration accounts are available!
- Students are eligible for a **FREE priority access account** to use resources funded by the Student Technology Fee via the Research Computing Club!



USER SUPPORT

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eScience Institute

Helpdesk

help@uw.edu

- Hyak, Tillicum, Kopah, Lolo, Cloud
- Research Computing Consulting
- Support Request form

Office Hours

- Drop-in or by Appointment
  - Calendar
- In-person and remote

Regular Training Opportunities

- Recorded
- Events Calendar
- We have a mailing list

Everything underlined is a link



RESEARCH COMPUTING CLUB

STF | Student Technology Fee Committee

Computing Resources

Hyak Klone:

- 26 nodes; 24 GPUs
- Priority job scheduling
- Computing storage

Tillicum:

- Apply for $250 in Tillicum computing credits

Cloud Credits Program

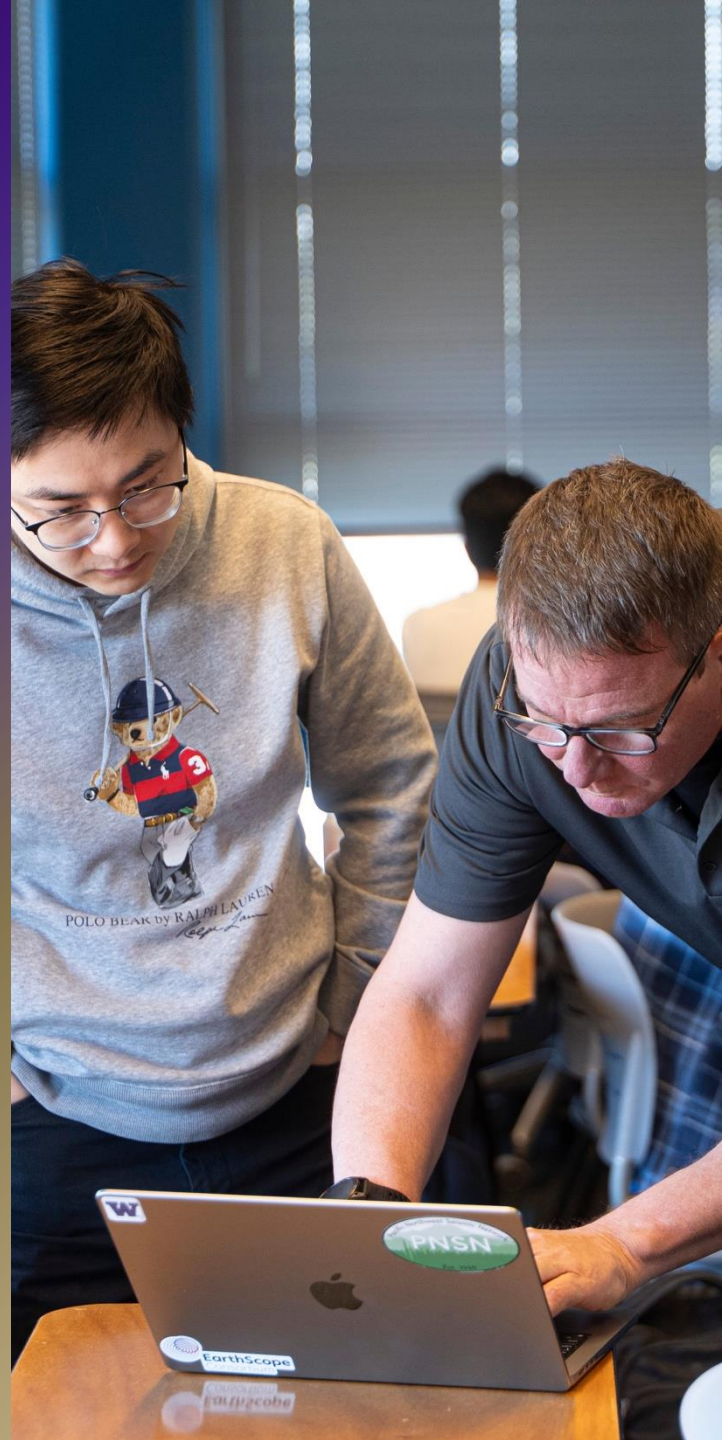
- Apply for $500 in AWS credits
- Guidance and troubleshooting

Slack Community

- Get help from your peers
- Get notified about events, jobs, and fellowships
- Get involved - recruiting new officers for Fall 2026!

Special Events

- Hackathon
- ... and more!



Documentation

www.hyak.uw.edu/

- Step by step guides to tools and software
- Monthly blog posts with updates
- Links to support and services

Learning Resources

www.hyak.uw.edu/learn

- Short How-to videos
- Full length tutorials

Everything underlined is a link



HOW DO I GET STARTED?

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eScience Institute

Regular Training Opportunities

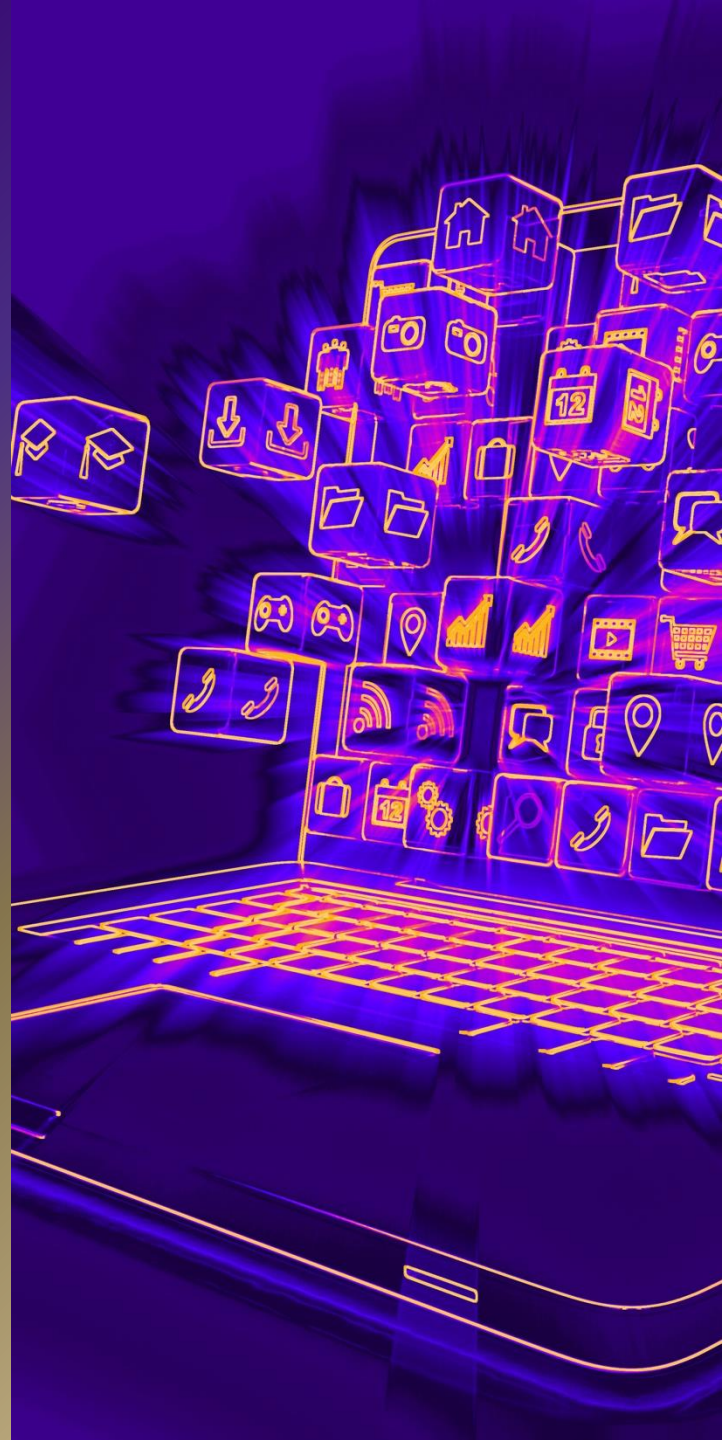
- Recorded
- Events Calendar
- RCC Slack Community

Fall 2026 Events

- Nov 5 10AM - Linux Fundamentals Allen Library Green Classroom A
- Nov 12 12PM - Accelerating Research Day HUB 145
- Nov 13 9AM - Research Computing Symposium
- Nov 18 10am – NVIDIA Office Hour (Virtual)

eScience Workshops

- Software Carpentry
- Pixel Proficiency - deep learning with images
- Python - your research assistant + weekly Seminars!
- Sign up for the Newsletter



Shared Community Policies

- Shared Research Environment
- Baseline Computing Environment
- No Root/Sudo Access
- Researcher Responsibility
- Hyak Team - Support, Training, and Documentation



Cluster Environment

Environment – Rocky Linux + Slurm

- Open OnDemand
  - Jupyter Notebooks
- Containerized environments (Apptainer runtime)
- LMOD modules - maintained and contributed modules
  - Flat vs. Hierarchy
- Python environments (conda module)



Learning Objectives

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By completing this tutorial, you'll learn how to:

- Discover your accounts and partitions on Klone
- Run interactive and batch jobs.
- Inspect job status, resource requests, and completed-job accounting.
- Write Bash workflows and run independent tasks with Slurm arrays.
- Choose resource sizes and understand checkpoint/preemption behavior.

Training Materials Repo:

https://github.com/UWrc/klone-slurm.git

Feedback Survey:

Slurm Job Scheduler Workshop



Learning Objectives

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NEED TO KNOW'S: training access period

- You will be able to complete whenever you want with your demonstration account (or other)
- Over the next ~6 days (ending Monday morning Nov 2) you will have priority access via UWIT (slurm account "uwit")
- This tutorial is designed as introductory concepts and practice followed by a "task" that combines multiple skills.

Training Materials Repo:

https://github.com/UWrc/klone-slurm.git

Feedback Survey:

Slurm Job Scheduler Workshop

Learn more – everything underlined is a link

UWIT - https://it.uw.edu/

Research Computing Services
- Tillicum: GPU-Accelerated Research Computing Platform
  - Intake Form
- Hyak: High-Performance Supercomputing Research Cluster
  - Pricing and Eligibility
  - Documentation
- Data Storage Services
  - Kopah S3 Object Storage
    - Pricing and Eligibility (cost calculator)
    - Documentation
  - Lolo Data Archive
    - Pricing and Eligibility
    - Documentation
- Cloud Computing
- Secure Computing
- Research Computing Consulting

Training and Events
- Hyak Blog
- UWIT Research Computing Calendar
- Office Hours
- eScience Newsletter (scroll for sign up)
- eScience Data Science and AI Accelerator
- Office of Research Calendar

Past Trainings
- Research Computing Tutorials & Video Library
- eScience YouTube Channel

New Hyak Users
- Free Demonstration Account
- Learning Resources
- Limitations of demonstration accounts
- Using free resources (Checkpoint)

Students
- Research Computing Club
- Student Hyak account
- Tillicum credits
- Cloud Credits
- Student Technology Fee

Other Resources
- **Join the UW AI Community of Practice** on MS Teams to get updates from UW-IT's AI team about events and join the discussion around AI in the news, society, and culture.
- UW Seattle Events Calendar

Special thanks to
- UW Office of Research
- eScience Institute